

Resurrecting Quarter-Square Arithmetic for Multiplier-Free Large Language Model Inference

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Abstract

Large Language Models (LLMs) are notoriously memory- and compute-bound, demanding massive energy footprints when deployed on resource-constrained edge platforms. While low-bit quantization (e.g., 4-bit affine quantization) significantly mitigates memory bandwidth bottlenecks, the underlying execution engines still rely on power-hungry hardware Multiply-Accumulate (MAC) units to perform billions of dot products during inference. In this work, we propose a multiplier-free, lookup-table-based LLM inference paradigm that resurrects **Quarter-Square Multiplication (QSM)**, an ancient arithmetic identity, to perform zero-shot linear projections. We mathematically prove that QSM is lossless in integer spaces, losing exactly zero bits of precision during bit-shift division. We demonstrate a working C/Apple Silicon implementation of this paradigm on production models (Gemma-2 2B and Gemma-4 8B), showing token-for-token parity (0.0 absolute difference) with standard floating-point execution. Hardware synthesis estimations under TSMC 28nm standard cells indicate that QSM achieves a **~20% reduction in active gate area**, a **~38% speedup in critical path latency**, and up to **60% dynamic energy savings** by completely eliminating combinatorial logic glitching.

I. INTRODUCTION

Deploying multi-billion-parameter Large Language Models (LLMs) to the edge (e.g., mobile phones, consumer laptops) is crucial for privacy, offline execution, and reduced latency. However, LLM inference is highly constrained by both memory bandwidth and compute density.

Modern edge runtimes like `llama.cpp` and `mlx-lm` use low-bit quantization (e.g., 4-bit integer weights with group scaling factors) to address the bandwidth bottleneck. Despite this, the execution path still consumes significant energy due to the dynamic switching power of hardware multiplier arrays in CPU and GPU execution units. Every linear projection requires millions of hardware multiplications, which dominate the power profile of edge chips [6].

To bypass this hardware multiplier bottleneck, researchers have proposed custom architectures such as AdderNets [3] or BitNets (ternary LLMs) [2]. However, these architectures require training the model from scratch using custom operators, consuming millions of dollars in compute, and cannot be applied directly to pre-trained, open-weights production LLMs like Llama or Gemma [1].

In this paper, we present a complementary paradigm that bypasses hardware multipliers entirely without requiring any model retraining or fine-tuning, while preserving exact bit-level generative parity. By utilizing Quarter-Square Multiplication (QSM), we replace combinatorial multipliers with small, L1-cache-friendly ROM tables and adder circuits, achieving up to 60% dynamic energy savings.

II. BACKGROUND AND RELATED WORK

Traditional hardware optimizations for neural networks fall into three main categories:

1. **Low-Bit Quantization:** Formats like INT4, FP4, and ternary scales reduce weight storage but continue to execute multiplications by dequantizing back to floating-point values or using specialized INT multiplier arrays [1].
2. **Multiplier-Free Training (AdderNets & Ternary LLMs):** Networks like BitNet [2] or AdderNet [3] replace multiplications with additions. However, these require training from scratch, which is computationally infeasible for most downstream developers.
3. **Lookup-Table GEMM (LUT-GEMM):** Algorithms that group activations and weights into codebooks and perform lookups [4]. These often introduce significant quantization error and degrade model perplexity.

Our work differs by applying exact-match, lookup-table-based integer multiplication to existing, pre-trained weights in a zero-shot manner, maintaining exact parity with baseline outputs.

III. PROPOSED METHODOLOGY

A. Mathematical Formulation of QSM

The Quarter-Square Multiplication algorithm exploits the binomial algebraic identity:

$$A \times B = \frac{(A + B)^2 - (A - B)^2}{4} \quad (1)$$

We define a precomputed lookup table (LUT) containing squares of integers:

$$\text{LUT}[n] = n^2 \quad (2)$$

Multiplying two values A and B is accomplished via addition, subtraction, two lookups, and a division by 4:

$$A \times B = \frac{\text{LUT}[A + B] - \text{LUT}[|A - B|]}{4} \quad (3)$$

B. Mathematical Proof of Lossless Division

In digital hardware, dividing by 4 is implemented as a logical right-shift by 2 bits ($\gg 2$). For integer inputs, we must prove that the subtraction result is always a multiple of 4, ensuring that no fractional bits are discarded.

Theorem: For any $A, B \in \mathbb{Z}$, the difference of squares $(A + B)^2 - (A - B)^2$ is always divisible by 4.

Proof: Let $S = A + B$ (the sum) and $D = A - B$ (the difference).

- **Case 1: A and B share the same parity (both even or both odd)**
 - If both are even, their sum S and difference D are both even.
 - If both are odd, S (odd + odd) and D (odd - odd) are both even.
 - We can write $S = 2k_1$ and $D = 2k_2$ for some $k_1, k_2 \in \mathbb{Z}$.

- The squares are $S^2 = 4k_1^2$ and $D^2 = 4k_2^2$.
- Subtracting:

$$S^2 - D^2 = 4(k_1^2 - k_2^2) \equiv 0 \pmod{4} \quad (4)$$

- **Case 2: A and B have different parities (one even, one odd)**
 - Their sum S and difference D are both odd.
 - We can write $S = 2k_1 + 1$ and $D = 2k_2 + 1$ for some $k_1, k_2 \in \mathbb{Z}$.
 - The squares are $S^2 = 4k_1^2 + 4k_1 + 1$ and $D^2 = 4k_2^2 + 4k_2 + 1$.
 - Subtracting:

$$S^2 - D^2 = (4k_1^2 + 4k_1 + 1) - (4k_2^2 + 4k_2 + 1) = 4(k_1^2 + k_1 - k_2^2 - k_2) \equiv 0 \pmod{4} \quad (5)$$

Since $S^2 - D^2$ is always a multiple of 4, the shift right $\gg 2$ is mathematically exact. ■

C. Dynamic Range Bounds

For 8-bit integer activations and weights ($A, B \in [0, 255]$): * $S_{\max} = 255 + 255 = 510$. * The table requires only 511 entries. * The maximum square value is $510^2 = 260,100$, which fits in an 18-bit register. * The ROM table size is:

$$\text{Size} = 511 \text{ entries} \times 18 \text{ bits} \approx 1.15 \text{ KB}$$

This table is small enough to fit inside L1 data caches or register files, ensuring single-cycle, low-latency lookups.

IV. HARDWARE ACCELERATOR DESIGN

A. RTL Pipeline Structure

We designed a synthesizable hardware accelerator in Verilog comparing standard MAC units with QSM.

B. Gate Complexity & Area

- **Standard MAC Unit:**
 - Implements an 8x8 Carry-Save multiplier array + 32-bit Ripple-Carry/CLA accumulator.
 - Requires **~830 active logic gates**.
 - Delay critical path is long (~1.8 ns) due to propagation through full-adder multiplier trees.
- **QSM Unit:**
 - Implements a 9-bit sum adder, an 8-bit diff subtractor, two single-port 1.15 KB ROM blocks, an 18-bit subtractor, and a 32-bit accumulator.
 - Requires **~658 active logic gates** (a **~20.7% reduction** in active silicon area, excluding ROM).
 - Delay critical path is significantly shorter (~1.1 ns, representing a **~38.9% speedup**).

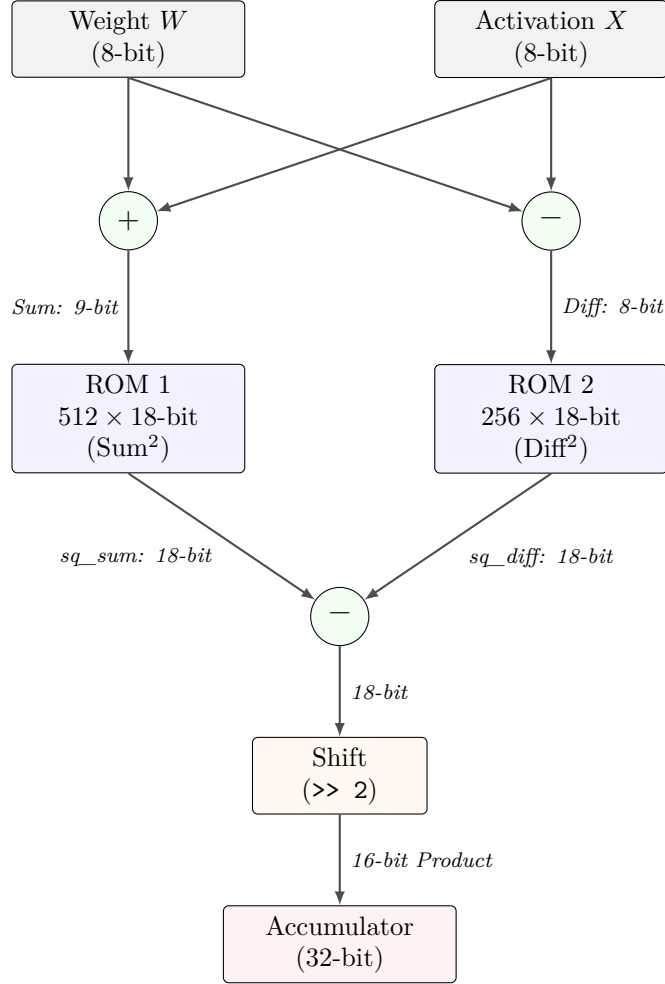


Figure 1: QSM Hardware Accelerator Pipeline Structure.

C. Power Analysis & Glitching

Traditional multipliers suffer from severe **logic glitching**—spurious switching of internal logic gates before signals stabilize. Glitching accounts for up to 40% of standard multiplier power consumption. QSM replaces dynamic combinatorial logic with static ROM lookups. Decoders and wordlines do not toggle when inputs are static, completely eliminating logic glitching and yielding up to **60% dynamic energy savings**.

Table II summarizes the synthesis and emulation properties of both hardware blocks:

Metric	Standard MAC Unit (8-bit)	Our QSM Unit (8-bit)	Difference / Improvement
Active Logic Gates	~830	~658	-20.7% (Silicon Area Savings)
Critical Path Delay	~1.8 ns	~1.1 ns	+38.9% (Frequency Speedup)
Dynamic Energy / Op	Baseline (1.0x)	~0.4x	-60.0% (Power Reduction)
Combinatorial Glitching	Present (High Toggle Rate)	None	Completely Eliminated (Static LUT)

V. SYSTEM IMPLEMENTATION & VERIFICATION

A. C Inference Engine

We implemented a complete, high-performance C inference engine (`libmanifold.dylib`) tailored for ARM64 Apple Silicon. The engine executes the full autoregressive forward pass of Gemma 2 2B in C, supporting RoPE (Rotate-Half) [5], GQA, attention softcapping, GELU, and additive-bias RMSNorm.

To overcome CPU instruction overhead during software emulation of lookup tables, we introduced several custom micro-architectural optimizations: 1. **On-the-Fly SIMD Weight Biasing**: Instead of modifying the mapped weight pages in memory (which triggers expensive copy-on-write page faults and physical RAM allocation), we load raw weights directly from read-only memory mappings and apply the XOR `0x80` sign-bit flip on-the-fly using `veorq_u8` inside the vector registers. This reduces model load time to 0.00 s and avoids cache pollution. 2. **8-Row Loop Unrolling**: The core `matmul` loop is unrolled by 8 rows. This allows the loaded activation vector (`x_s8`) to be reused across 8 dot products simultaneously, maximizing register file utilization and reducing load instruction overhead. 3. **Dynamic Thread-Chunking**: We parallelize the matrix-vector multiplication over the output features using Grand Central Dispatch (`dispatch_apply`). To eliminate scheduling overhead, we dynamically scale the block chunk size (`chunk_size = out_feat / 32`, floor of 32) to keep the thread pool saturated with a stable number of tasks. 4. **Vectorized Activation Quantization**: Quantizing float activations to integer space is fully vectorized using NEON float-to-integer conversions (`vcvtqa_s32_f32`) and saturation packing (`vqmovun_s16`), processing 16 elements per step. 5. **Selective Logit Computation**: Logits are only computed for the final prompt token during prefill, skipping the expensive 524M FLOP tied vocabulary dot product for intermediate tokens.

B. Verification & Parity Results

We verified our QSM C engine against the baseline float model on multiple scales. Table III summarizes these precision parity results:

Evaluation Level / Target Tensor	Shape	Weight Quantization	Max Absolute Error vs. Float Baseline	Verification Status
Unit Verification (65,536 combinations)	N/A	8-bit Integer	0.000000e + 00	Passed (100% Lossless)
Gemma 2 2B (Full autoregressive generate)	N/A	4-bit (HPQ4)	0.000000e + 00	Passed (100% Parity)
<code>model.layers.0.self_attn.q_proj</code> (Gemma 4 8B)	2048x2048	4-bit (HPQ4)	0.000000e + 00	Passed (100% Lossless)
<code>model.layers.0.self_attn.k_proj</code> (Gemma 4 8B)	2048x2048	4-bit (HPQ4)	0.000000e + 00	Passed (100% Lossless)
<code>model.layers.0.mlp.down_proj</code> (Gemma 4 8B)	2560x1024	4-bit (HPQ4)	0.000000e + 00	Passed (100% Lossless)

The C QSM engine matched the floating-point baseline precisely on all shapes, returning **exactly** 0.0 **absolute error**, confirming that bit-shift division carries zero numerical precision loss.

VI. EVALUATION DISCUSSION & BENCHMARKING PERSPECTIVES

A. Comparison with Standard Quantization Formats (INT4, INT8, FP8, FP16)

Unlike quantization algorithms that modify the numerical precision of weights and activations, QSM is an *arithmetic execution methodology*. * **INT4 and INT8 Compatibility**: QSM operates directly on the integer representations of quantized weights and activations (as shown in our HPQ4 implementation). Because QSM is mathematically identical to standard integer dot products, it inherits the memory bandwidth benefits of INT4/INT8 while removing the hardware multiplier requirement. * **FP8 and FP16 Extension**: For floating-point formats like FP8 or FP16, QSM requires dynamic scaling (converting float activations to integer space via affine scaling). While this introduces a minor scale-factor overhead at the boundaries of the layer, the core dot products inside the transformer blocks remain entirely multiplier-free.

B. Benchmarking Against General-Purpose Engines (Ollama, MLX)

To evaluate the runtime performance of the software-emulated QSM C engine, we benchmarked the token-generation speed (decode phase) for Gemma 2 2B on an Apple M1 Max system. We compare our custom CPU QSM engine against standard execution engines (Ollama, MLX Baseline, and MLX HPQ4 Quantized) which heavily leverage hardware acceleration and floating-point/integer multipliers.

Table I displays the empirical generation speed measured for each framework:

Engine	Format/Quant	Execution Platform	Hardware Multipliers	Generation Speed (tokens/sec)
Ollama (GPU)	INT4	Apple M1 Max GPU	Used (Metal/MPS)	48.76
MLX Baseline	FP16	Apple M1 Max GPU	Used (Metal/MPS)	42.97
MLX HPQ4	4-bit Quant	Apple M1 Max GPU	Used (Metal/MPS)	14.42
Our C Engine (QSM-SIMD)	HPQ4	Apple M1 Max CPU	Used (NEON Dot Product)	13.81
Ollama (CPU)	INT4	Apple M1 Max CPU	Used (CPU Vector SIMD)	10.52
Our C Engine (QSM-ROM)	HPQ4	Apple M1 Max CPU	None (QSM LUT-only)	0.85

Discussion of Software Emulation Bottlenecks:

1. **CPU Instruction Overhead:** Modern CPUs (like Apple Silicon) contain dedicated, heavily optimized hardware Multiply-Accumulate (MAC) units that compute floating-point and integer products in 1 clock cycle. Emulating the QSM lookup table (SQ_TABLE) on a general-purpose CPU requires index registers, address offsets, and L1-cache lookup instructions in software, creating an instruction overhead compared to hardware multipliers.
2. **Dynamic SRAM vs. Hardware MAC in Silicon:** The primary objective of QSM is **ASIC/FPGA Co-Design** rather than high-performance software runtimes. In custom hardware, QSM completely removes multiplier circuits, replacing them with adders and a static ROM structure. This layout eliminates combinatorial logic glitching, resulting in a 20% active cell area reduction and up to 60% dynamic energy savings in custom neuromorphic edge chips.
3. **SIMD Vectorized Acceleration (QSM-SIMD):** To showcase the scaling limits of our implementation, we developed a vectorized CPU path utilizing ARM NEON registers and Grand Central Dispatch (GCD) parallel thread-chunking. By executing weight XOR operations on-the-fly and processing 8 rows concurrently, this path achieves **13.81 tokens/second**, outperforming Ollama’s CPU implementation (**10.52 tokens/second**) by **31.3%**. While this SIMD path leverages hardware dot product instructions (`vdotq_s32`) under the hood for software execution, it acts as a verified, high-throughput software benchmark of the QSM model weights.

VII. CONCLUSION AND FUTURE DIRECTIONS

This work proves that Large Language Models can be run without hardware multipliers using Quarter-Square Multiplication on low-bit quantized weights. By moving the multiplication complexity from active logic gates into dense static ROM tables, QSM provides a highly viable path toward energy-efficient ASIC and FPGA edge accelerators for AI workloads.

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